final-fixed-methodology

Finalized Approach B — Placement-Conditioned Sampling and Joint X/Y Corrective-Sequence Learning

Latent Causal Infeasibility for Long-Horizon Task and Motion Planning

5 November 2026

Abstract

This document freezes the final scientific architecture for Approach B. A separate placement-only sampler supplies object–destination candidates that are predicted to be robot-placeable, conditional on the object having been grasped. A branch-specific, bounded sequence search composes these candidates; the existing root-consistent, prefix-conditioned X/Y Boltzmann model predicts whether each complete ordered sequence can physically execute (X) and whether its hypothetical final object arrangement restores the original target task (Y). Search minimizes a combined joint-success and correction-cost objective. Sampler queries may be repeated after hypothetical scene edits, while the X/Y model always evaluates entire sequences relative to the original root. The sampler never determines the target-repair solution, the X/Y model never generates destinations or robot trajectories, and no explicit causal graph or learned intermediate robot configuration is introduced. The 5,000-scene 2D and 5,000-scene 3D corpora are retained as the scene basis; new individual placement labels are collected as versioned sidecars on these same roots. This is a normative methodology, not a claim that its new sampler integration or physical evaluations have already been completed.

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DOCUMENT STATUS — FINAL SCIENTIFIC ARCHITECTURE / IMPLEMENTATION PENDING. This is the authoritative intended Approach B architecture from 5 November 2026. "Final" fixes the separation of responsibilities, X/Y event semantics, model family, inference contract, data provenance, and evaluation claims. It does **not** assert that the sampler, dynamic search integration, 5,000-scene label sidecars, or 3D experiments have been completed. The conditional VAE versus flow-matching choice, analytical sampling implementations, sample/search budgets, and confidence-calibration mechanism remain *implementation and ablation choices within this frozen architecture*, not invitations to redesign Approach B.

Reader's one-paragraph summary. The robot receives an initially blocked target task. For each movable obstacle, a sampler proposes destination locations that appear physically placeable **assuming that object has been grasped**. These proposals do not identify blockers, do not decide the number or ordering of moves, and do not resolve the target task. A bounded search constructs ordered combinations of the proposed relocations, and can ask the sampler for additional placements in hypothetical edited scenes as a branch grows. The existing X/Y model evaluates each *entire ordered sequence* from the *original* robot and object state: X predicts sequential physical executability with no intermediate reset, and Y predicts success of the original target task in the resulting hypothetical object arrangement. The selector favors high **joint X-and-Y success** and low correction costs. The actual robot controller remains external. No test-time planning/IK verifier is required to admit every sampled destination.

1. Final architectural commitments

1.1 The scientific question

Given an original scene S_0, an initial robot configuration q_0, a specified target task T and its goal g_T, and a collection of movable obstacles, can a system identify a short, executable ordered sequence of obstacle relocations that makes T feasible, **without receiving a solution-informed catalogue of repairs**?

The question has two distinct statistical parts: (i) does a candidate ordering of relocations actually execute with the available robot skills; and (ii) does its final object arrangement make the original target manipulation feasible? The central Approach B claim concerns the *joint learned evaluation and cost-aware selection* of candidate sequences. The sampler supplies a useful candidate domain but is not a substitute for that joint evaluation.

1.2 What is definitively fixed

1. **Two responsibility-separated learned components.** A placement-only sampler supplies destinations; a separate main model scores whole ordered corrective sequences through X and Y. They need not share weights or training objectives.
2. **Sampler input/output.** Input: hypothetical object scene, selected movable object, support/placement domain, and frozen robot capability convention. Output: a finite set of proposed destinations and optional calibrated placement-feasibility confidence. The sampler is conditional on the object having been grasped; it does not need to decide whether the object can be picked.
3. **No destination invention by the main model.** The X/Y model scores only object–action–destination tuples supplied by the sampler/sequence constructor. It chooses *among* destinations, objects, lengths and orders, never generates the destination itself.
4. **No explicit intermediate-state learner.** Order-dependent physical executability is learned implicitly from ordered-prefix features and supervised complete-sequence outcomes. The model does not generate trajectories, IK solutions, joint-space waypoints or an explicit learned $q_1, q_2, \ldots$ state sequence.
5. **Two binary outcome types only.** X_r is the budgeted physical executability of the complete ordered corrective prefix plus terminal connection to the target reference state; $Y_{s(r)}$ is the budgeted feasibility of the original target task from the reference state in the hypothetical edited scene. Success requires **both**.
6. **Root-consistent joint Boltzmann model.** Preserve F2 ordered-prefix features, F3 hypothetical-scene features, the narrowly defined F5 quadratic X–Y interaction in the coupled arm, omission queries, exact observed-data marginal likelihood, and optional travel regression. The source-verified implementation supplies coupled, uncoupled and direct controls.
7. **Dynamic, branch-local candidate generation.** A sampler may be queried again after applying a partial sequence to a *hypothetical copy* of the scene. Previously chosen actions are frozen within that branch; other branches remain intact. The number and timing of such queries are compute-budget controls.
8. **Joint success-and-cost selection.** Rank complete stopping candidates by predicted joint $P(X = 1, Y = 1)$ and explicitly chosen length, object-displacement and robot-travel costs. A low-Y *partial* prefix may be extended; it is not thereby declared a successful repair.
9. **No explicit causal DAG or blocker-identity output.** Causal significance is tested through interventions, omissions, changed destinations and order contrasts. The model is not claimed to recover a unique causal graph.
10. **No privileged proposed-repair catalogue.** Candidate generation must not read intended repairs, certified blocker sets, generator construction hints, Y/X labels, teacher outcomes or successful witnesses at test time. Small, sound static legality rules are permissible and must be disclosed. Analytical and learned samplers are compared independently of task-repair success.
11. **No per-sample online IK/motion-planner veto.** Sampled placements are high-confidence proposals, not certified executable endpoints. Offline label generation and independent evaluation may and should use the pinned physical teacher; actual execution remains a separate external system.
12. **Existing scene corpora remain the basis.** The stated 5,000 2D and 5,000 3D scene roots are used, subject to audit of what actually exists. New individual-placement and intermediate-scene labels are added to versioned sidecars of those roots rather than creating a separate randomized scene dataset.

These commitments distinguish the architecture from both the historical construction-informed P20 catalogue and the universal but nearly unfiltered P26 action universe.

1.3 Choices deliberately not frozen

The precise sampler family is not yet selected: conditional VAE and conditional flow matching are the primary learned candidates; Sobol/stratified sampling and the independently specified swept-volume sampler are the principal analytical comparisons. Network width/depth, inference steps, sample count per object, branch-expansion budget, probability calibration, and sampling schedules are chosen by development-only experiments. None changes the interface above.

Scope caution. "Finalized" applies to the intended method. It cannot pre-commit the empirical conclusion that coupling, a learned sampler, or this architecture will outperform its simpler controls. Negative experimental outcomes must remain reportable.

2. Precise problem and representation

2.1 Root, objects, tasks and action grammar

A root is

$$\mathcal{R} = (S_0, q_0, q_{\text{ref}}, T, g_T, \mathcal{O}, \mathcal{F}, \Pi, \mathcal{B}),$$

where $\mathcal{O}$ contains movable objects, $\mathcal{F}$ fixed objects/structure, Π denotes declared robot/grasp/motion capability and teacher policy, and $\mathcal{B}$ denotes relevant execution budgets. The scene includes object identities, poses, shapes and supported resting regions. The target action is fixed within a root; the purpose of rearrangement is to make this *same* target action feasible.

An individual corrective action is

$$a_i = (o_i, \text{relocate}, d_i), \qquad o_i \in \mathcal{O},$$

where d_i is an explicit resting destination (2D cell or 3D placement pose, as appropriate). An ordered correction is

$$r = (a_1, \dots, a_m), \qquad 0 \le m \le K_{\text{search}}.$$

The empty sequence is the no-op. The current 2D teacher grammar moves an object at most once and permits a finite relocation horizon; the current universal-search study uses $K_{\text{search}} = 4$. The earlier finalized teacher allows up to six corrective relocations, subject to the pinned total-action contract. A 3D experiment can set a compatible finite horizon (for example K0–K3 evaluation) without changing the method. The study must report actual allowed action depth, not silently conflate the search budget with the teacher maximum.

2.2 Initial and hypothetical scene states

For a prefix $r_{1:t}$, let

$$S^{r_{1:t}} = \text{Edit}(S_0, r_{1:t})$$

be the *hypothetical* edited object scene resulting from applying its specified destinations. It is a deterministic geometrical bookkeeping operation, **not** a witnessed robot execution. Every unchanged object retains its original pose, and the target and fixed structure remain unchanged.

The original robot state q_0 and target-reference state q_{ref} are held fixed when scoring a complete ordered candidate. Hypothetical edits do not produce a learned or assumed achieved robot state q_t. This is intentional. The physical teacher, not the learned representation, carries the true achieved robot configuration during offline X evaluation.

2.3 A placement is not a successful correction

A valid resting pose can be statically legal yet robot-unplaceable. A robot-placeable destination can be useless for the target. Two individually placeable destinations can be incompatible when executed sequentially. Thus three notions remain separate:

- **Static eligibility:** object fits, is supported, is not placed on fixed/occupied geometry, and respects the declared workspace convention.
- **Sampler placement plausibility:** estimated feasibility of placing a grasped object at a destination under a pinned grasp/transport capability protocol.
- **Sequence success:** $X_r = 1$ *and* $Y_{s(r)} = 1$ for the full candidate correction, not the sampler's confidence.

Only the first two are responsibilities of the proposal interface. In particular, no independent target-clearance oracle is allowed to cut the sampler down to a hidden set of already-correct repairs.

3. Why the earlier implementation was insufficient

3.1 Historical small catalogue (P20–P25)

The historical `bxy_p20_k34.candidates` generator builds variants of an intended geometry-informed correction: destination changes, reversed order, a few permutations, prefixes, omissions, wrong destinations and interference examples. Its upstream construction uses access geometry and source fields such as `construction.intended`, `construction.options`, `neutral_cells`, `q_ref_footprint` and object roles. It is therefore **solution-informed candidate construction**, even though the resulting individual actions are not all physically verified. It is not the proposed final sampler.

That catalogue provided useful supervision and diagnostic contrasts, but it made evaluation easier than genuinely independent destination selection. The model learned to rank members of a small, biased set; it was not yet shown to rank broadly sampled robot-placeable destinations.

3.2 Universal domain and bounded search (P26–P28)

The separate `approach-b-universal-search` branch implements a lazy domain over every movable object, every currently empty legal cell, every order, and every sequence length through $K_{\text{search}} = 4$. Median development-root domain size is about 1.37×10^9 ordered sequences versus about 11 historical catalogue entries. These numbers describe an action-grammar universe, **not** 1.37 billion distinct endpoint scenes or 1.37 billion physically feasible sequences.

P27 supplies a completeness-in-principle bounded search with exact accounted coverage and a fairness fallback. P28 integrates the frozen X/Y ensembles without score drift (maximum numerical difference approximately 2.2×10^{-16} against legacy scoring on the checked examples). Yet only about 3,100 candidates per development scene were scored, and 3–5 of eight certifiable cases selected a correction shorter than the independently certified necessary length. Models also displayed elevated ensemble disagreement and off-catalogue displacement.

The result is not that the X/Y equations are conceptually wrong. It is that (i) the historic proposal distribution was narrow, (ii) unrestricted combinatorics can be extremely large, (iii) the frozen model is not demonstrably reliable off that distribution, and (iv) finite search exploration can be too shallow. Neither P28's predictions nor its analytic certificates are a substitute for new physically verified, sampler-conditioned evaluation.

3.3 New fixed resolution

The final method interposes a **placement-only sampler** between scene geometry and sequence construction. Unlike the old construction-informed catalogue, it does not know the target's intended repair. Unlike unrestricted enumeration, it does not send virtually every empty object–destination combination to the X/Y model. It offers a finite set of likely robot-placeable endpoints; X/Y remains solely responsible for ranking their ordered combinations by complete sequential executability and restoration of the original target task.

This division reduces proposal complexity while preserving the specific scientific problem of joint corrective-sequence learning. It does **not** guarantee that a finite sampler will include the necessary repair; coverage is an explicit experiment.

4. End-to-end architecture

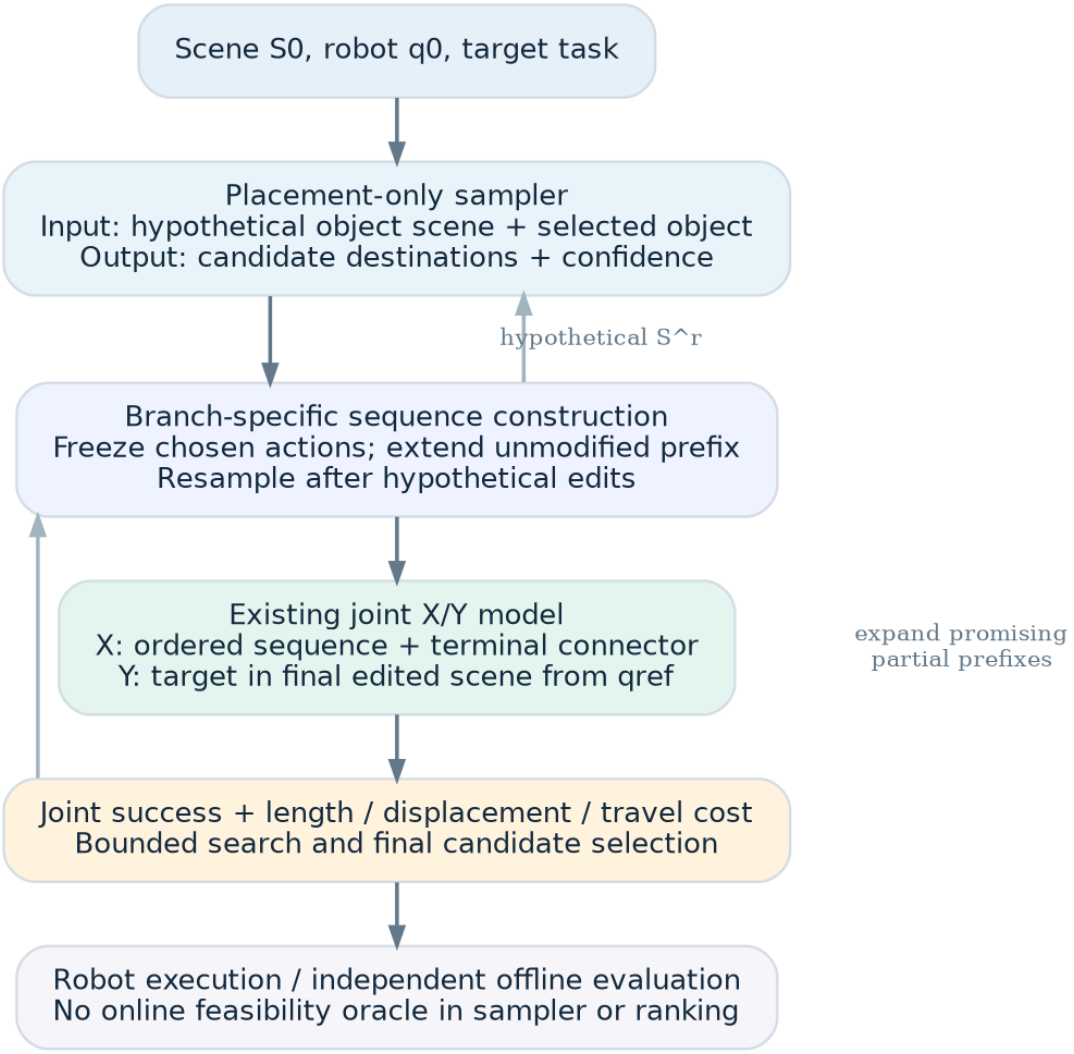


Figure 1: Figure 1. Frozen component boundaries of Approach B. The feedback arrows indicate bounded branch expansion, not online model retraining.

4.1 The three principal runtime stages

Stage I — Placement proposals. For each object considered at a search node, sample plausible destination poses. The sampler is evaluated with only lawful scene, object, robot-capability and support information, not with a hidden target repair label. A learned sampler can be cVAE or flow matching; an analytical sampler can supply equivalent typed proposals.

Stage II — Candidate composition. The search constructs complete stopping candidates from already chosen object/destination pairs. At an expanded branch, the chosen prefix is fixed, the hypothetical edited scene is formed, and the sampler can generate new placements for objects not yet relocated. Other branches retain their own prefixes and candidate sets.

Stage III — Joint outcome evaluation and cost-based selection. Given the **original root** and a complete ordered candidate r, X/Y predicts the joint probability of physically executing r and restoring T. A selector compares complete stopping candidates using this probability plus costs. The external controller ultimately executes a selected candidate; sampler and scorer do not output trajectories.

4.2 What never crosses the component boundary

The sampler does not read target success labels, candidate X/Y scores from the current root as ground truth, constructed blocker lists, or certified minimum necessary relocations. The X/Y scorer does not read sampler training labels, sampled probability values as proof of X, teacher observations about current test candidates, or intended-repair construction metadata. The search may use the sampler's scores to order sampling or retain diversity, but it must not silently multiply them into the calibrated joint-success probability without an explicitly defined and tested model.

A sampler placement-confidence score and the X/Y joint-success probability describe **different conditional events**. They are not interchangeable and can be correlated. Multiplying them naively would usually double-count aspects of placement feasibility already subsumed by X.

5. Placement-only sampler: authoritative interface

5.1 Precise role

The sampler answers the following question:

> Assuming a selected object o has been grasped using the robot's permitted manipulation capabilities, what destination poses are plausible for placing/releasing it in the current object arrangement?

It does **not** need to answer whether the object can presently be picked, whether the robot can arrive at the grasp after previous actions, whether the resulting arrangement helps the target, or whether a whole sequence ends at the required target-reference robot state. Those belong to the complete-sequence X and Y predictors.

For scene S and object o, the output is a **finite** candidate collection

$$\widehat{\mathcal{D}}_\phi(S, o; M) = \{(d_i, c_i, \mu_i)\}_{i=1}^{M},$$

where d_i is a legal destination pose; c_i is a declared score/confidence; and μ_i records sampling provenance or metadata (source method, support, object, seed, score meaning). The number M is a runtime budget. A generative sampler cannot literally return every continuous 3D location; "all possible destinations" means it may sample from the full permitted domain without a hidden intended-repair whitelist, with coverage measured empirically.

5.2 What does placement success mean?

Define an offline placement trial under pinned capability protocol Π_{place}. The experimental condition is that the object has been successfully grasped or presented in one of the allowed post-grasp states. A placement succeeds if the robot can transport the held object to the requested pose, place/release it under the declared support/contact rules, and satisfy the declared local completion condition. It **need not** return to q_{ref} and **need not** perform the target task.

This is an operational, robot-policy- and budget-relative event $V_{\text{place}}(S, o, d; \Pi_{\text{place}})$. In grasp-agnostic deployment, a pinned rule aggregates across the permitted post-grasp conditions (for example, successful under at least one tested allowed grasp; or success rate over a fixed grasp distribution). These two aggregations are **not mathematically identical**; one must be declared before labelling. No particular grasp token or explicit q_t is required as a runtime input to the main X/Y model.

The sampler therefore supplies **plausible placements conditional on a grasp convention**, not an assertion that every grasp can reach each location. If B is not currently pickable, it may still be proposed: complete X rejects B-first if needed, while A-then-B can remain possible.

5.3 Why geometric placeability and robot placeability differ

A stable nonoverlapping resting pose is a necessary geometric condition but not sufficient for an articulated robot to place an object there. The robot might lack a collision-free carry path, safe approach/release orientation, or adequate workspace. This motivates learning from *actual placement trials* rather than treating empty cells as positive placement labels.

Conversely, the sampler should not solve the full ordered robot task. It needs only a **placement-conditioned proposal**, for which a finite probability of failure is acceptable. Any low-level motion-planning or IK used to produce *offline training labels* is separate from its test-time function.

5.4 Density is not feasibility probability

Two quantities must be kept distinct:

$$p_\phi(d \mid S, o, V_{\text{place}} = 1) \qquad \text{versus} \qquad c_\phi(S, o, d) \approx P(V_{\text{place}} = 1 \mid S, o, d).$$

A cVAE or flow-matching generator trained primarily on successful placements estimates a distribution **among successful placements**. Sample frequency, latent likelihood or flow density is not automatically the probability that a given placement will execute. If the interface claims calibrated c_i, it must be trained or calibrated against independent positive and negative placement trials with a declared intervention/sampling distribution. A separate calibration head *inside the sampler module* is an allowed implementation detail; it is **not** an extra causal, target-repair or intermediate-motion model.

If only uncalibrated generative scores are available, they must be named `proposal_score` or `density`, not `p_feasible`. Fixed-score or unscored candidate sets remain valid for sampler comparison, with offline precision measured independently.

5.5 Minimal legality versus privileged feasibility filtering

The sampler interface can reject destinations that violate the declared action grammar: a missing support surface, overlap with an existing object, fixed-geometry penetration, an out-of-bounds coordinate, a nonexistent object, or a prohibited repeated relocation. These checks prevent meaningless actions and are performed identically for all sampler baselines.

It must **not** prefilter using a target-specific successful motion, intended blocker set, successful repair library selected from this test root, downstream X/Y ground truth, certified K, or a full test-time IK/motion-planner sweep. Hard analytic rules should be documented and strictly narrower than the physical feasibility question being learned.

5.6 The two useful responsibilities of a sampler score

The score can help (i) allocate a fixed proposal budget across diverse destinations, and (ii) expose its own calibration/precision for the sampler-alone study. It is **not a replacement for X**. When the final sequence is scored, X already accounts for whether those placements are executable together with compatible grasps, approaches, carries, inter-action transitions and the terminal connector.

6. Sampler alternatives and frozen experimental comparison

The architecture does not select a winner by intuition. Every sampler must implement the **same input/output contract** and be tested under matched scene splits, object coverage, legal placement domain and proposal budgets.

6.1 Conditional variational autoencoder (cVAE)

Let $z \sim \mathcal{N}(0, I)$. A conditional decoder outputs a proposed destination pose

$$d = G_\phi(z; S, o), \qquad z \sim p(z).$$

The cVAE encoder $q_\psi(z \mid S, o, d)$ is trained on recorded *successful* placement examples. A standard negative evidence-lower-bound objective is

$$\mathcal{L}_{\text{cVAE}} = \mathbb{E}_{q_\psi}\big[-\log p_\phi(d \mid z, S, o)\big] + \beta_{\text{KL}} D_{\text{KL}}(q_\psi(z \mid S, o, d) \| p(z)).$$

For 2D, the output could be a categorical distribution over legal lattice cells or continuous coordinates later mapped to legal cells. For 3D, the output is a placement pose with explicit position, orientation

and support representation. Circular angles, rigid poses and support constraints require appropriate parameterization, not a naïve Cartesian mean across distinct orientation modes.

A trained cVAE draws diverse destinations in one or a few decoder passes. It does not automatically label them feasible. Positive/negative trials are used to evaluate and, if needed, calibrate its confidence separately.

6.2 Conditional flow matching

Let x_0 come from a simple source distribution and x_1 be a recorded successful placement. A conditional vector field $v_\phi(x, t; S, o)$ learns to transport noise toward the distribution of successful destination poses. Under a specified interpolant x_t and target velocity u_t, a representative objective is

$$\mathcal{L}_{\text{FM}}(\phi) = \mathbb{E}_{S,o,x_0,x_1,t}\left[\left\|v_\phi(x_t, t; S, o) - u_t\right\|_2^2\right].$$

At inference, a numerical ODE solver integrates the learned field from noise to candidate destination. The solver step count, conditional representation and rejection of statically illegal outputs determine actual runtime. Flow matching may be expressive for multi-modal continuous 3D placements, but greater expressiveness does not by itself guarantee better feasible-action coverage or better end-to-end repairs.

For a small finite 2D lattice, the continuous output must be mapped to legal cells and duplicates removed; a method should not gain apparent coverage from repeated samples of one destination.

6.3 Sobol / stratified analytical sampler

A Sobol or stratified sampler proposes destinations with balanced coverage of the legal placement domain. In 2D, sample or order legal cells directly. In 3D, select support surfaces, then sample local translation/orientation coordinates and project only to statically valid resting poses. It has **no learned individual-placement-feasibility confidence** unless paired with a separately declared scoring procedure.

Sobol is important as a non-learning control: it tests whether simple even coverage is enough for the X/Y ranking model, or whether a learned sampler actually concentrates effort in useful robot-placeable regions. Comparison must be based on equal numbers of *unique eligible proposals*, not merely equal raw draws when one method creates many duplicates.

6.4 Swept-volume-based analytical sampler

The swept-volume alternative remains deliberately **method-unspecified** pending the user's revised design. The common interface allows it to propose destinations using its own independently declared geometric features and any offline cache. The final report must disclose whether its swept volumes represent generic robot manipulation capability, historical motions, hypothetical individual carries, or trajectories associated with a specific target repair. These have different levels of privileged task information.

Until that source is fixed, the methodology makes **no unsupported claim** that swept-volume sampling is cheap, complete, unbiased, or physically valid. It receives no special target-repair oracle unavailable to other methods, and it must not receive successful test-root witnesses or test outcomes through a lookup table.

6.5 Fair sampler-alone comparison

Evaluate each sampler *before* the main X/Y scorer enters the experiment. Compare at the same object budget and proposal counts $M \in \{M_1, M_2, M_3\}$ selected on development data. The primary measurements are:

- **Verified placement precision:** fraction of independently attempted sampled destinations for which the pinned offline placement teacher actually succeeds.
- **Placement coverage:** how broadly the sample set recovers independently evaluated positive destinations or supported successful placement regions, subject to the resolution of the reference evaluation.
- **Diversity and effective sample size:** unique destinations, distinct regions, coverage of near/far locations, and duplication/collapse rate.

- **Repair-preservation recall:** fraction of scenes in which at least one full verified target-restoring repair can be formed from the sampler's action proposals within the declared sequence budget. This requires independent downstream verification for *evaluation*, not test-time sampler filtering.
- **Computational cost:** sampling latency, memory, number of model/lookup calls, and cost of repeated queries on hypothetical scenes.
- **Confidence quality (if claimed):** Brier score, reliability diagram and risk/coverage curves for c_ϕ, using the same evaluation trial distribution and respecting class imbalance.

A sampler can achieve high placement precision yet fail repair preservation by repeatedly proposing safe but useless destinations. That failure should be reported rather than hidden by high X/Y classification accuracy on its restricted pool.

7. Training data: reuse the existing 5,000 + 5,000 scenes

7.1 Fixed scene basis

The user reports separate **5,000-scene 2D** and **5,000-scene 3D** datasets currently being constructed, with 3D variants from the no-repair case through K3. These are the intended scene corpora for the final study. Their contents, completeness, unique root counts, splits and current label fields are not asserted here as independently audited facts.

The known or expected contents primarily concern full corrective sequences and complete repair outcomes. Such a record is **not sufficient by itself** to label every individual object–destination placement. A failed AB sequence could have failed grasping A, carrying A, the A-to-B transition, placing B, or returning to q_{ref}; it does not establish whether A's or B's individual destination is placeable under the sampler convention.

Final data decision: Do not create a new independent randomized/planned scene corpus just to train the sampler. Instead, run additional *offline placement-labeling experiments on the existing scene roots* and store them in independently versioned sidecars. This is extra trial collection within the same scene corpus, not a new scene-generation campaign.

7.2 A placement-trial record

For each selected scene root and individual placement trial, record at minimum:

Field	Meaning
`root_id`, `scene_hash`, `domain`	Immutable scene identity and 2D/3D domain
`prefix_key`, `hypothetical_scene_hash`	Which edited object configuration was presented, if any
`object_id`, `source_pose`, `destination_pose`	Candidate object and explicit destination
`support_id`, `action_contract`	Resting surface/contact and permitted place/release skill
`grasp_protocol_id`	Declared conditional-on-grasp trial convention; not an X/Y latent variable
`attempted`, `observed`, `result`	Attempted status and observed success/failure, or missing
`negative_kind`, `failure_phase`	Certified, unsuccessful within budget, timeout, collision, invalid grasp condition, etc.
`teacher_hash`, `robot_hash`, `budget_hash`	Reproducible supervision policy identity
`witness`, `runtime`, `seed`	Optional successful trace, cost and exact randomization provenance

The exact JSON schema can remain minimal, but its semantics cannot be ambiguous. A genuine placement label should isolate the placement/carry/release event under the declared post-grasp condition; it should **not** silently inherit the complete X result with its terminal connector and preceding picks.

A repeated measurement with the same scene/object/destination but different grasp or teacher policy is not automatically a duplicate of the same event. Such trials must preserve their policy identity before any prescribed aggregation into grasp-agnostic labels.

7.3 How to collect labels on these same roots

1. Freeze root-level training, validation and test partitions *before* generating new sidecars. Keep all edited descendants, rerenderings, alternative placements and reversed-order sequences derived from the same root within its partition.
2. Within a training root, propose a broad set of object–destination trials independent of the construction-intended repair. Cover different source objects, distance bands, directions, support surfaces, congestion levels and legal cells/poses. Use a frozen exploratory proposal distribution; do not make the label set entirely dependent on the sampler being trained.
3. Run the pinned offline teacher in a **placement-only** query protocol conditioned on the declared grasp-success convention. Save positives, negatives, censored outcomes and motion witnesses. Record static invalidity separately when trials are screened before teacher execution.
4. Include **hypothetically edited scenes from the same roots** so the sampler is exposed to configurations where moving A changes feasible placement regions for B. Record the exact parent prefix and edit; do not pretend an infeasible prefix was physically executed.
5. Sample some outcomes from unsuccessful, neutral and non-solution edits, not just intended-successful repair traces. A sampler trained exclusively on successful intended repairs risks reproducing the historical proposal bias.
6. Stop labelling according to a *predeclared coverage/compute policy*, not when a preferred sampler obtains favorable metrics. Keep attempted/censored counts explicit.
7. Train each sampler on the same allowed training-root sidecars. Freeze model selection on validation roots. Lock final test roots before comparing sampler precision or full corrections.

The offline teacher can employ IK, contact checks and continuous motion search to obtain labels. The architecture prohibits **online candidate-vetoing by a full teacher call**, not the physical supervision needed to learn an honest placement distribution.

7.4 How much labelling is needed?

The existing scene count alone does not determine the number of required placement trials. In a 2D scene with five movable objects and 55 free lattice cells, there can be about $5 \times 55 = 275$ distinct individual object–destination pairs, before changes in hypothetical prefixes. Across 5,000 such scenes this would give about 1.375 million pairs, but that number is only an illustrative *action inventory*; it is not the actual size of the user's datasets or a mandate to label every pair.

The labelling budget is selected after measuring success prevalence, diversity of successful regions, teacher expense, and learning-curve saturation. Report labelled outcomes per scene, per object, per distance bin, per domain and per intermediate-prefix depth. Positive examples must not be so rare that a generator can only imitate one local placement mode.

7.5 Data bias, leakage and provenance

The earlier dataset construction may favor specific blocker destinations and short movements. Retaining its scenes is acceptable; **pretending its original catalogue provides unbiased placements is not**. The additional labelling distribution must be recorded and broad enough to test novel destinations within the same scene families. An appropriately labeled dataset can still have generalization limits outside those families; disclose them.

Never mix a root's hypothetical descendants across train/test. Never use a successful X/Y test witness to expand the test-time sampler catalogue. Store root hashes and both query/teacher hashes. Reused full-sequence labels must match the exact X/Y definitions; otherwise keep them as auxiliary historical outcomes, not silently relabel them.

7.6 Shared 2D/3D semantics, separate evidence

The goal is one *methodological interface* across 2D and 3D, not necessarily one single set of trained weights. A 2D cell and a 3D stable pose differ in dimensionality and contact constraints. Use domain-adapted input encoders/decoders while keeping the same roles, target labels and evaluation questions. Cross-domain transfer is an optional later experiment, not a prerequisite for claiming the final architecture was implemented in each domain.

8. Dynamic resampling on hypothetical edited scenes

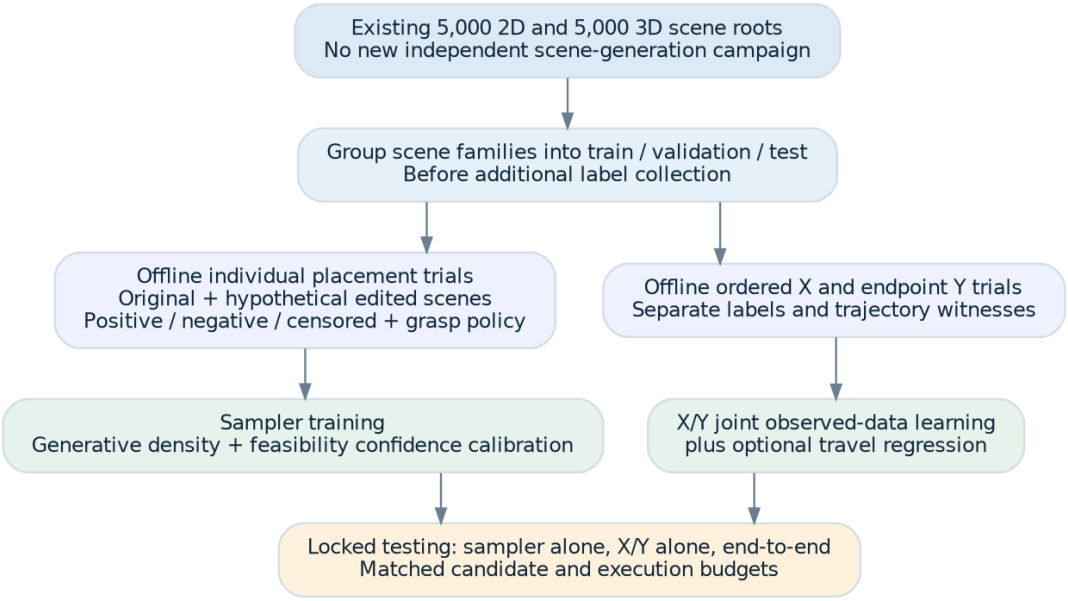


Figure 2: Figure 2. Offline sidecars for sampler placement outcomes and X/Y sequence outcomes. The two supervision tracks share root identities but not label definitions.

8.1 Why a one-time static action pool can be insufficient

If the initial scene blocks access to B, a sampler that only proposes fully pick-and-place executable initial actions might never propose B. Moving A can open B's grasp region. A fixed initial candidate set would then miss a feasible A-then-B repair.

Our sampler has now been defined as **placement-conditioned**, so it may still propose B even when B is not initially graspable. This reduces—but does not eliminate—the motivation for dynamic resampling. The available *placement destinations* for B can still change after A vacates or occupies space. Dynamic queries therefore remain useful even if pick feasibility is left completely to X.

8.2 Branch-local state, not a single overwritten global catalogue

For a search branch $r_{1:t}$, maintain the tuple

$$\mathcal{N}_t = (r_{1:t},\ S^{r_{1:t}},\ \widehat{\mathcal{A}}_t,\ \hat{p}_{\text{succ}}(r_{1:t}),\ C(r_{1:t}),\ \text{search metadata}),$$

where $\widehat{\mathcal{A}}_t$ is the newly available per-object placement proposal set for that *branch*. To expand $\mathcal{N}_t$:

1. Freeze the object identities and destinations already chosen in its prefix.
2. Form $S^{r_{1:t}}$ by deterministic object-pose edits only.
3. Choose each remaining eligible movable object; query the sampler for candidate destinations in this hypothetical scene.
4. Apply basic static eligibility; deduplicate identical actions and endpoint scenes.
5. Append each proposed action, producing children $r_{1:t+1}$.
6. Score those full ordered prefixes from the original root using the existing X/Y predictor.
7. Put children on the search frontier with their scores, coverage bookkeeping and provenance.

The sibling branch $r'_{1:t}$ retains *its own* edited scene and proposals; there is no global mutation of S_0 or a destructive reset of candidates. The procedure supports a beam, best-first, or fair best-first search under declared budgets. Exact search scheduling is an engineering choice, not a third learned component.

8.3 Root-relative X/Y scoring after every sample extension

Suppose the branch is $[A \to P]$ and the sampler on S^A proposes $B \to Q_2$. Then the main model receives

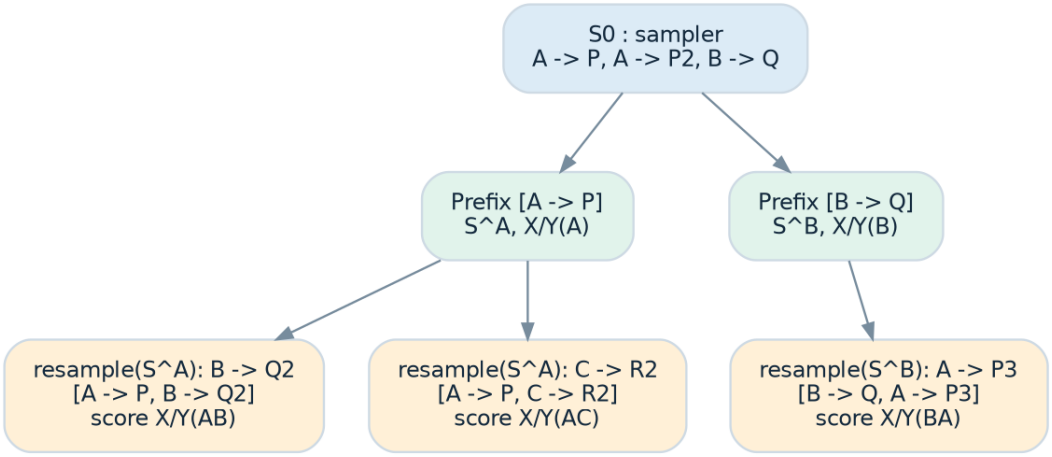


Figure 3: Figure 3. Each branch freezes its chosen prefix and obtains updated placement proposals from its own hypothetical object scene.

$$(S_0, q_0,\ r = [A \to P, B \to Q_2]),$$

not merely $(S^A, B \to Q_2)$ as a fresh independent physical root. The sequence encoder processes A then B, so it can represent order interactions. The Y encoder processes the final edited arrangement S^{AB} from q_{ref}. This retains exact semantic consistency with training.

Reusing a cached X/Y result for a previously seen ordered signature is allowed. Two different orders with identical final object poses reuse Y but have distinct X nodes. The sampler may also cache responses keyed by `(hypothetical_scene_hash, object_id, support_domain, sampler_version, sampling_seed/budget)`.

8.4 Does each sampler call force an X/Y call?

No one-to-one synchronization is required. The sampler may create many placements or operate on several hypothetical scenes in a batch; the search assembles newly formed sequences and scores them in subsequent X/Y batches. Previously computed X/Y scores and shared endpoint Y predictions are reused. **Neither trained model is updated during inference.** Repeated invocation is a forward pass, not retraining.

The search may postpone scoring/expanding some proposals if compute is bounded, but it must distinguish unexamined sequences from verified failures. A method limited by proposal or search budget cannot claim no solution exists merely because it did not sample one.

8.5 A low-Y prefix is not a dead branch

$Y_{S^A} = 0$ may be entirely correct if A is only the first of three necessary obstacle removals. The main model optimizes X and Y **for complete stopping candidates**. It does not impose a monotonic Y increase after every action and does not demand $Y_{S^A} = 1$ before A may be extended. Similarly, $X_A = 0$ due solely to a failed terminal connector does not prove that AB is impossible, because the connector is required only if the robot **stops** after A; AB continues directly from the achieved post-A state.

Heuristic priorities can delay low-scoring branches, but must not interpret low predicted probability, an observed nonterminal Y=0, or an observed terminal-connector failure as a sound subtree-impossibility proof. Explicitly state the bounded-search policy and its possible incompleteness under finite budgets.

8.6 Expected compute behavior

Let $M_t(o)$ be the number of sampled destinations for a remaining object at a particular node. Its branching factor is roughly $\sum_{o \in \mathcal{O}_{\text{remaining}}} M_t(o)$, rather than the number of all legal cells times all ob-

jects. The total search tree remains combinatorial; even modest M and K can grow quickly. A sample budget, model-score batch size, active-frontier cap and fairness policy are therefore necessary.

The overhead of cVAE may be dominated by its encoder/decoder; flow matching may require many ODE steps; Sobol and cached swept-volume methods have different amortized costs. **Do not assume dynamic sampling is cheap before measuring it** on both 2D and 3D. The method only requires that this cost be accounted for, not that it be negligible.

9. Main X/Y outcome semantics: unchanged and authoritative

9.1 X: complete ordered physical execution

For each distinct ordered corrective prefix r, define the observed outcome

$$X_r \in \{0, 1\}.$$

The physical query begins at the **original** scene S_0 and robot state q_0. It attempts the relocations in the specified order using the pinned grasp/motion policy and finite budgets. The achieved robot joint state after each relocation is carried into the next; **there is no return to home or q_{ref} between A and B**. After the *last* relocation, if treating that prefix as a stopping correction, the X query includes a validated connector to q_{ref} in the edited scene. The connector condition ensures a successful X witness can be concatenated with the independently evaluated Y target witness.

Operationally, $X_r = 1$ means successful completion of all these requirements under the declared teacher policy and budgets. $X_r = 0$ means an attempted query did not meet them under that policy; it does **not** establish that the sequence is impossible under every imaginable controller or unlimited computation. A query not attempted is *missing*, not a negative label.

The terminal connector is a stop-condition, not a between-action reset. This is why a partial $X_A = 0$ may coexist with a longer feasible $X_{AB} = 1$ if A's physical relocation succeeded but its stopping-point connector failed. That case must be preserved in tests and in search.

9.2 Y: hypothetical target restoration

For each distinct edited endpoint scene $s = S^r$, define

$$Y_s \in \{0, 1\}.$$

Y is evaluated by independently attempting **the original target task** in edited scene s, from the same fixed q_{ref}, with no previous relocation trajectory assumed to have occurred. The target task and fixed structures do not change. This Y experiment is meaningful even if the corresponding corrective sequence could not be physically executed, so $(X, Y) = (0, 1)$ is a valid outcome.

Y depends on the final object arrangement and frozen task-reference convention, **not on the order by which that arrangement was proposed**. Thus two permutations producing identical object poses share one Y variable; their X outcomes can differ.

9.3 The four outcomes are all meaningful

X	Y	Meaning
0	0	Ordered correction failed within budget, and target still infeasible in the hypothetical scene.
0	1	Hypothetical final arrangement would restore the target, but this ordered correction could not execute and connect within budget.

X	Y	Meaning
1	0	Ordered correction and connector execute, but target remains infeasible.
1	1	Composable successful correction and target under the pinned teacher protocol.

The candidate must score highly on the **joint (1,1) event**, not merely on one marginal.

9.4 Order effects without an explicit intermediate robot variable

The X encoder receives ordered corrective action tokens with predecessor-aware interactions and the original scene/robot context. It is supervised by complete ordered-sequence physical outcomes. Therefore it can learn that a placement after A changes access or motion options for B, even though it never predicts q_A or constructs the intervening trajectory.

This is the intended abstraction. It is a *predictor of feasible sequential outcomes*, not a learned dynamics simulator. It captures order effects statistically at the level required for repair ranking. The offline teacher retains full physical evidence and exposes failure phases for interpretation and auditing.

9.5 Same endpoint, different order: why the factorization matters

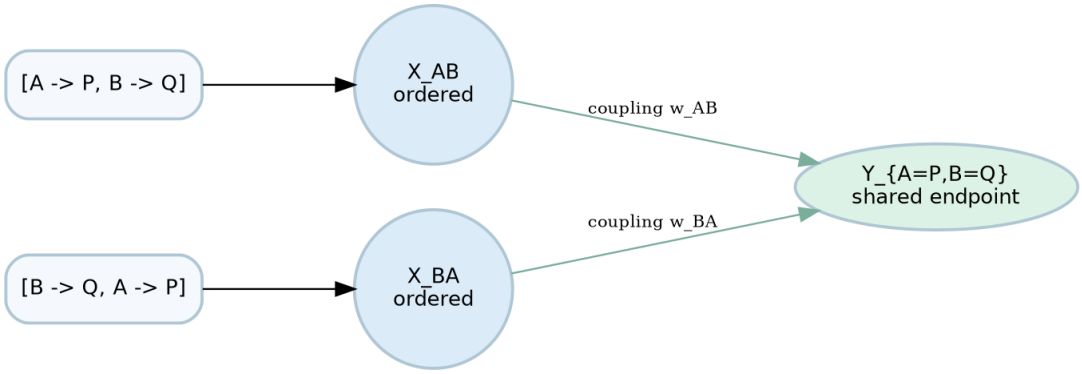


Figure 4: Figure 4. AB and BA have different X variables, but share Y because they specify the same final object poses. This is a dependency graph, not a causal DAG.

Suppose $r_1 = [A \to P, B \to Q]$ and $r_2 = [B \to Q, A \to P]$. Then

$$S^{r_1} = S^{r_2}, \qquad Y_{s(r_1)} = Y_{s(r_2)}, \qquad X_{r_1} \not\equiv X_{r_2}.$$

The last expression means X outcomes are allowed to differ; it does not assert that every reversed order must differ. An experiment should include cases where ordering matters and cases where it does not.

10. Main model: exact joint Boltzmann formulation

10.1 Neural representations and coefficients

Let $v_s = f_{\text{scene}}(S^r, q_{\text{ref}}, T, g_T)$ encode a hypothetical endpoint scene and original target. Let $g_r = f_{\text{order}}(S_0, q_0, a_1, \dots, a_m)$ encode the complete ordered correction using predecessor-aware interactions. Let c be frozen root/task context.

The relevant learned scalar coefficients are

$$b_s = h_Y(v_s, c), \qquad a_r = h_X(g_r, v_s, c), \qquad w_r = h_{XY}(g_r, v_s, c).$$

The uncoupled control sets $w_r = 0$; the coupled primary model learns it. These quantities are **energy/logit coefficients**, not obstacle IDs, trajectory costs or inferred numbers of blockers. The trained modules can also predict a travel readout $\widehat{J}_r$. In the current `bxy_p11.Model` implementation, coupled D, uncoupled E and DIRECT are distinct comparison arms, with actual source-level details documented in the implementation reference.

10.2 Pairwise energy and the correct normalization

A convenient raw local energy for a corresponding X/Y outcome pair is

$$\widetilde{E}_r(x, y) = -a_r x - b_{s(r)} y - w_r xy, \qquad x, y \in \{0, 1\}.$$

Important: this *raw* energy is not itself the complete normalized root model when several X candidates share one Y endpoint. Naively multiplying local pair factors and globally normalizing would allow adding more candidates to change Y's scene-only marginal—a catalogue-size artifact that the finalized methodology explicitly rejects.

Instead, define a normalized scene-first factorization, for the unique endpoint set $\mathcal{S}$ and distinct ordered-query set $\mathcal{Q}$ of one root:

$$P_\theta(\mathbf{y}_\mathcal{S}, \mathbf{x}_\mathcal{Q} \mid \mathcal{R}) = \prod_{s \in \mathcal{S}} P_\theta^Y(y_s \mid \mathcal{R}) \prod_{r \in \mathcal{Q}} P_\theta^X(x_r \mid y_{s(r)}, \mathcal{R}, r).$$

For temperature $\tau > 0$ (currently pinned to 1 in the source formulation), the conditional factors are

$$P_\theta^Y(Y_s = 1) = \sigma(b_s/\tau),$$

$$P_\theta^X(X_r = 1 \mid Y_{s(r)} = y) = \sigma((a_r + w_r y)/\tau)\,.$$

The normalized root energy is

$$\begin{aligned} E_\theta^{\text{root}}(\mathbf{x}, \mathbf{y}) = &\sum_{s \in \mathcal{S}} \left[-b_s y_s + \tau \log\left(1 + e^{b_s/\tau}\right)\right] \\ &+ \sum_{r \in \mathcal{Q}} \left[-(a_r + w_r y_{s(r)}) x_r + \tau \log\left(1 + e^{(a_r + w_r y_{s(r)})/\tau}\right)\right]. \end{aligned}$$

This energy has an implicit unit partition function under its included normalizers. The X-dependent normalizer is Y-dependent; omitting it invalidates the claimed root-consistent marginal. Distinct X alternatives sharing the same Y do not change the learned $P(Y_s = 1)$ simply because more alternatives were proposed.

10.3 What the main model actually outputs

The coupled model can return for each candidate:

$$\begin{aligned} p_Y(r) &= P(Y_{s(r)} = 1) = \sigma(b_{s(r)}/\tau), \\ p_{X|Y}(r) &= P(X_r = 1 \mid Y_{s(r)} = 1) = \sigma((a_r + w_r)/\tau), \\ p_{\text{succ}}(r) &= P(X_r = 1, Y_{s(r)} = 1) = p_Y(r)\, p_{X|Y}(r). \end{aligned}$$

It can also report the marginal

$$P(X_r = 1) = p_Y(r)\, \sigma((a_r + w_r)/\tau) + (1 - p_Y(r))\, \sigma(a_r/\tau),$$

which is **not** the same quantity as $p_{X|Y}$ when $w_r \neq 0$. At test time, the model never conditions on an undisclosed teacher-revealed Y label. A direct-success baseline instead outputs p_{succ} without the separate X/Y decomposition.

The empty repair $r = \emptyset$ has $X_\emptyset = 1$ by the declared target-reference convention, so its success probability is simply $P(Y_0 = 1)$. A no-op gate can prevent unnecessary correction if the original target already looks feasible.

10.4 Learning energies versus choosing actions

During training, optimize the neural parameters θ that determine a_r, b_s, w_r (and $\widehat{J}_r$), using observed outcomes and the objective in Section 11.

During inference, keep those parameters frozen and search over proposed discrete corrective sequences r. Compute the normalized $p_{\text{succ}}(r)$ and rank candidate sequences by a decision cost. This is **not** gradient-descent optimization of the raw $\widetilde{E}_r$ with respect to object destinations. The destinations already come from the sampler.

This distinction resolves a common confusion: the learned energy distribution provides **probabilities of physical/target outcomes**; the selector uses those probabilities to evaluate **candidate corrections**.

11. Training the existing X/Y model and the sampler

11.1 Two datasets, one shared scene-root provenance

The sampler sidecar and X/Y sidecar can be generated from the same original scene corpus, but their supervision differs:

- **Sampler records:** conditional *individual placement* outcomes V_{place}, destination poses, prescribed grasp convention, policy and failed phase.
- **X records:** complete ordered relocation-prefix outcomes with sequential achieved robot state carried forward and terminal connector to q_{ref}.
- **Y records:** original target-task outcome in a distinct hypothetical edited endpoint scene from fixed q_{ref}.
- **Travel records:** measured cost only when an appropriate successful complete physical witness is available.

A sampler label must not be substituted for X. A successful X trace can contain placement-level evidence, but it must be parsed according to the actual failed phase and grasp convention before being reused as a sampler label. A full-repair success does not label every other object/destination pair in the scene.

11.2 Observed-data likelihood

Let $\mathcal{O}_Y$ and $\mathcal{O}_X$ denote the actually observed unique Y and X queries for one root. An unattempted query is missing, not zero. The main training loss averages the negative log marginal likelihood of **observed** evidence per root:

$$\mathcal{L}_{\text{joint}}(\theta) = -\frac{1}{N_{\text{roots}}} \sum_{c=1}^{N_{\text{roots}}} \log P_\theta(D_c \mid \mathcal{R}_c).$$

For each unique hypothetical endpoint Y, sum over its allowed Y states; for each observed child X, include $P(X_r = x_r \mid Y_s)$. Unobserved X variables sum to 1 and contribute no fabricated binary target. The formula is evaluated exactly for this lean graph by per-Y grouping, rather than enumerating every joint outcome state.

A distinct X ordered query may appear in several catalogues or omission examples, but counts once per recorded root. Multiple orderings that lead to the same edited scene share a single Y observation and coefficient. Averaging by root prevents scenes with many sampled candidates from silently dominating the loss.

Teacher failures and censoring

Under the earlier fixed teacher convention, an *attempted* budget-limited X or Y query that fails to verify is operationally labelled 0, with negative kind and resource consumption retained. An unattempted query is not labelled. For newly collected sampler trials, maintain the same explicit provenance and distinguish experimentally declared operational failure from censored/no-attempt data. Do not call a finite-budget negative a proof of physical impossibility.

11.3 Travel supervision and the final ranking cost

Let J be a versioned measure of joint-space travel over a successful complete witness: approach, transport, release, inter-action free motion, terminal connector and target action, with one pinned normalization. Let $\widehat{J}_\theta(r)$ be its learned prediction. A supported auxiliary objective is

$$\mathcal{L}_J = \frac{1}{N_J} \sum_{w \in \mathcal{W}_{\text{complete}}} \text{Huber}\Big(\log(1+\widehat{J}_\theta(r_w)), \log(1+J(w))\Big) .$$

Then the unchanged Approach B learning objective is

$$\boxed{\mathcal{L}_B = \mathcal{L}_{\text{joint}} + \lambda_J \mathcal{L}_J.}$$

λ_J is a training hyperparameter; a failed action does not receive an invented large travel label. A cost head predicts a property of the pinned witness-provider policy, **not** the globally shortest possible motion path.

At inference, for an admissible **complete stopping sequence** r, the selector uses

$$\boxed{C(r) = -\log \max(p_{\text{succ}}(r), \epsilon) + \alpha m(r) + \beta C_d(r) + \gamma \widehat{J}_\theta(r).}$$

$m(r)$ is the exact action count, $C_d(r)$ deterministic total object displacement, and α, β, γ validation-frozen *selection* coefficients. They are not the neural energy coefficients and need not equal λ_J. The P28 common-rule example uses a no-op threshold 0.5, $\alpha = 0.3$, $\beta = 0$, $\gamma = 0$; that is a historical evaluation rule, not a universal physical law or a mandatory coefficient choice for the new sampler-conditioned dataset.

Minimizing $C(r)$ jointly considers **both X and Y**, because p_{succ} is their joint success probability. A high X/low Y candidate does not become a successful repair. A low-Y intermediate prefix can nevertheless remain available for later extension.

11.4 Sampler objectives and calibration are separate

The cVAE and flow-matching likelihood/velocity losses concern the distribution of conditional individual-placement destinations. They are **not terms in** $\mathcal{L}_B$ unless an explicitly predeclared joint training experiment is added as an ablation; the finalized primary architecture treats sampler training and X/Y learning separately.

If calibrated placement confidence is required, train or calibrate a placement-feasibility readout against independent labelled positive/negative trials. A separate sampler confidence loss may be binary cross-entropy or a proper scoring rule, with sampling-distribution bias accounted for. This remains **internal to the sampler** and does not alter X/Y's two-outcome factorization.

11.5 Distribution matching without information leakage

Training X/Y exclusively on the historical geometry-informed catalogue and testing it on a new cVAE/flow-generated candidate distribution is a known distribution-shift risk, directly illustrated by P28. The final development protocol must therefore collect or select **candidate-diverse X/Y supervision on the same original scene roots**, without target-oracle-based test-time filtering. The model family and equations remain fixed while its training coverage broadens.

For clean attribution, compare: historical frozen X/Y on the new sampler distribution; retrained X/Y on diverse training-root candidates; and DIRECT and G controls trained on matched evidence. Never

use final test outcomes or P24 confirmation outcomes to tune sampler or ranking thresholds. Matched supervision is essential before claiming the factorization helps.

12. Worked example: placement proposals, repeated calls, X/Y and energy

This example is **illustrative**, not a physical measurement. Let A and B initially obstruct the target T. A can be grasped and placed at P. B is initially hard to grasp, but moving A to P may open it. The sampler's responsibility is not to decide whether B can be grasped; it is to propose plausible destinations assuming grasp success.

12.1 First sampler call at the original root

The sampler returns, say, $\{A \to P, A \to P_2, B \to Q\}$. It may return B even if B cannot be picked initially. The search forms the individual stopping candidates $[A \to P]$, $[A \to P_2]$, $[B \to Q]$, and the no-op. For each candidate, X/Y predicts joint success, not the sampler's conditional placeability.

Assume $[A \to P]$ has $P(Y = 1) = 0.10$ and $P(X = 1 \mid Y = 1) = 0.95$. Then

$$p_{\text{succ}}([A \to P]) = 0.095.$$

The model correctly suggests that moving A alone is unlikely to restore the target. However, the branch may still be expanded, because adding B could solve it.

12.2 Hypothetically edit the scene; resample

Create S^A by moving only A's **object pose** to P. Do not execute a robot trajectory or infer an achieved q_A. Query the sampler again for B and other remaining objects, using S^A. Suppose it now returns B destinations Q_1 and Q_2; Q_2 may be newly available because A vacated a nearby cell. The branch extends to $[A \to P, B \to Q_2]$.

The main X/Y scorer receives **the original** (S_0, q_0) and the full ordered two-action candidate. Its Y representation uses S^{AB}, and its X representation encodes A then B. It need not learn any explicit intermediate joint state. Another branch that starts with B remains available and can be evaluated separately.

12.3 Compare the two-action orders

Suppose the resulting arrangements of AB and BA are identical and the model estimates the following:

Full candidate	p_Y	$p_{X\|Y}$	Joint success	Interpretation
$[A \to P]$	0.10	0.95	0.095	Executable-looking but insufficient alone.
$[A \to P, B \to Q]$	0.90	0.80	0.720	Promising complete correction.
$[B \to Q, A \to P]$	0.90	0.15	0.135	Same target-restoring scene, bad order.

The shared 0.90 Y probability reflects the common endpoint. X differentiates physical sequence feasibility. No separate intermediate-state predictor or causal blocker variable is required.

With $\alpha = 0.3$, $\beta = \gamma = 0$ for illustration, the costs are

$$C(A) \approx 2.65, \qquad C(AB) \approx 0.93, \qquad C(BA) \approx 2.60.$$

Thus AB wins among these complete candidates. **The intermediate A was not accepted as a successful solution.** It was only retained as an extendable prefix while its terminal success probability was low.

12.4 Three-action case and nonmonotonic prefix outcomes

If A, B and C are all required to reopen T, the correct physical outcome can be

$$Y_0 = Y_A = Y_{AB} = 0, \qquad Y_{ABC} = 1.$$

This is neither a contradiction nor a learning failure. A system that insists on $Y_A > 0$ or discards a branch whenever the prefix target remains infeasible would be incorrect for precisely this problem class. The search must distinguish *potential to extend* from *success if stopped now*.

12.5 What is cached?

- Placement proposals: by hypothetical scene, object, support domain, sampler configuration and sampling seed/budget.
- X: by the original root and **ordered sequence** signature; AB and BA are different.
- Y: by original root, target-reference condition and **final edited scene** signature; AB and BA can share one result.
- Costs: exact length and object displacement by sequence; predicted travel by ordered sequence; teacher witnesses strictly offline/evaluation only.

The candidate pool is therefore an expanding *union of branch-local proposals*, not a single list overwritten after each hypothetical edit.

13. Final inference algorithm and stopping semantics

13.1 Inputs, outputs and invariants

Inputs: original root $\mathcal{R}$, trained sampler G_ϕ, trained X/Y model F_θ, declared object/placement legality, relocation horizon K, per-object proposal budget M, search/compute budgets and validation-frozen selector hyperparameters.

Outputs: either a selected ordered corrective sequence r^* with its predicted p_Y, $p_{X|Y}$, p_{succ}, cost, candidate provenance and search-coverage status; or an explicit bounded-search abstention/unknown. A low probability is not an impossibility certificate. A budget-limited failure to find a candidate is not proof that no repair exists.

Invariants: original root and reference robot state remain immutable; hypothetical scenes are object-only edited copies; chosen actions remain frozen inside each branch; the same object is not relocated twice under the current action grammar; X/Y inference sees the whole original-root sequence; Y is deduplicated by endpoint; no current-root teacher label enters sampler/search/scorer; all methods obey identical static eligibility constraints.

13.2 Algorithm (methodological pseudocode)

```
Algorithm: FINAL-FIXED APPROACH B / Bounded Dynamic Placement Sampling
Input: root R=(S0,q0,qref,T), sampler Gphi, joint scorer Ftheta,
       horizon K, proposal budget M, search budget B, selection rule C
Output: best predicted complete correction r* or UNKNOWN/ABSTAIN

1  frontier <- {empty prefix ()}; results <- {}; caches <- empty
2  Score no-op Y0, store r=() as a possible stopping candidate.
3  while frontier nonempty and search budget remains:
4      Select an expandable prefix r from frontier by frozen
       best-first/fair scheduling; DO NOT demand Y(r)=1.
5      S^r <- apply the prefix's object-pose edits to original S0.
```

```
6       for each eligible unmoved object o:
7           D <- sample placements Gphi(S^r, o, budget M)
8           filter only the shared legal support/overlap/action grammar
9           for each unique destination d in D:
10              r_child <- append (o, relocate, d) to r
11              Record branch-local sampler scores and provenance.
12              Add r_child to score batch if not already cached.
13      For each new r_child in the score batch:
14          Evaluate Ftheta(S0, q0, qref, T, FULL ORDERED r_child)
15          Read pY(endpoint), pX_given_Y(order), and p_success.
16          Compute candidate stopping cost C(r_child), add to results.
17          If |r_child| < K, retain it for possible further expansion
            under the declared search scheduling policy.
18      Account for examined, deferred and excluded-by-static-rule nodes.
19  Return cheapest admitted complete stopping candidate observed,
    or UNKNOWN/ABSTAIN with search/proposal budget statistics.
```

"Complete stopping candidate" here means that the currently accumulated prefix is evaluated **as if we stopped relocating now** and attempted the target. It does not mean the candidate necessarily has length K or is feasible. Every prefix can be a potential final correction; prefixes with low Y may also serve as stepping stones to later corrections.

The pseudocode deliberately does **not** specify cVAE versus flow matching, a particular heuristic for expanding prefixes, or whether sampler probabilities are used to allocate draws. These are controlled implementation choices. It also does not claim to have preserved the exact completeness guarantee of P27 after introducing a finite generative sampler: **with a restricted, stochastic sampler, completeness holds only relative to generated proposals, not to every legal placement in the full domain**.

13.3 Selection, abstention and physical execution

The no-op is always considered. If $P(Y_0 = 1)$ exceeds the frozen no-op threshold under the declared rule, choose no-op unless a separate, preregistered safety constraint forbids it. Otherwise rank the **examined** nonempty candidates by $C(r)$ and a declared feasibility threshold η. When no candidate qualifies, abstain or defer to a separately declared baseline; do not invent a successful correction.

The chosen r^* is an **ordered high-level corrective specification**, not a low-level executable trajectory. A robot-side controller, motion planner or manipulation policy must still realize its pick/carry/place motions in practice. The final method does not call that controller to certify every sampled destination **before** X/Y ranking. During research evaluation, independent teacher verification of selected candidates remains essential to measure true success.

13.4 Why energy minimization and branch exploration are not contradictory

There are two distinct decisions:

- **Current-stop decision:** If the robot stopped after prefix r, how good a complete correction would r be? This is answered by minimizing $C(r)$, which uses **both X and Y**.
- **Expand decision:** Is it worth generating children of r that might become better complete corrections? This is a search-control decision. A low $P(Y_r = 1)$ does not entail low $P(Y_{r+a} = 1)$ for all extensions a.

It is therefore coherent to assign A a poor stopping cost and still explore AB. One must not silently interpret current-stop cost as a valid lower bound on all descendants. The method does not impose such a bound.

14. 2D and 3D embodiment contracts

14.1 2D: articulated robot, discrete resting positions

The 2D experimental embodiment is a physically modeled articulated planar arm on the declared 8-by-8 object-resting lattice, with nonzero-width links, gripper, object attachment/release and continuous robot configuration space. Resting locations are discrete cells, but robot motions are **not** grid walks. A legal empty destination can still be unreachable due to collision or motion constraints.

Sampler outputs are cells (and, if the action convention requires it, permitted discrete placement orientation/mode). The X teacher evaluates continuous robot motion between ordered relocations without intermediate reset; Y independently tests the original target task in the edited scene from q_{ref}. The sample budget is per object and per hypothetical scene, not a substitute for a correct physical teacher.

14.2 3D: support-constrained object placement

The 3D experiment extends the same semantics to a real articulated 3D manipulation embodiment—such as the planned xArm/cupboard-style scene family—with object shapes, supporting shelf/table surfaces, gripper clearance and constrained upright or other declared resting orientations. The sampler outputs complete target placement poses or a support-surface parameterization sufficient to reconstruct them.

In 3D, a destination may be spatially empty but unplaceable because of object height, shelf aperture, attached-object sweep, wrist orientation, contact/settling stability or collision with surrounding geometry. These are the reasons for collecting **placement-specific teacher outcomes**, rather than labelling arbitrary free 3D coordinates positive.

The chosen 3D grasp/transport protocol and support pose family must be pinned and reported. The main X/Y factors retain exactly the same *logical* meaning as in 2D, while encoders, pose features, grasp capabilities and physical teachers may be domain-specific. No claim of shared 2D-to-3D network weights is required.

14.3 Scope boundaries common to both domains

The corrected architecture addresses **high-level corrective relocation and target restoration**. It does not require novel contact-rich manipulation, pushing, object deformation, unknown-object discovery, open-world language planning, a generic motion-generating policy, or proof of universal physical infeasibility. Such extensions are separate future research questions. Source objects and goal definitions remain explicit in the evaluated root.

15. Evaluation, comparisons and falsification

15.1 Three evaluation layers; do not conflate them

Layer A — Sampler alone: Is a destination likely to be individually placeable under the declared conditional-on-grasp protocol? Measure offline placement precision, coverage, diversity, latency and confidence calibration.

Layer B — Main X/Y model: Given matched sets of full ordered candidates and matched physical evidence, can the model predict joint feasibility and discriminate ordering and endpoint interventions? Measure joint NLL, Brier, calibration, X/Y decomposition, order contrasts and cost errors. This is a *fixed-candidate* comparison, so sampler quality is not credited as X/Y predictive skill.

Layer C — End-to-end corrective search: Does the complete sampler + sequence construction + frozen X/Y + bounded selector recover physically verified repairs within finite budgets and with reasonable move count and travel? Measure successful repair selection, abstention, verification calls, regret within a declared verified candidate pool, repairs absent from the historical catalogue, missed repair cases and resource use.

The same training/validation/test root policy applies to all three layers. An improvement in Layer C alone does not prove which component caused it; use the ablations below.

15.2 Sampler comparison matrix

Compare conditional VAE, conditional flow matching, Sobol/stratified, and the separately specified swept-volume method. Use matched destination legality, robot/tool capability description, object identities, support constraints, candidate limits and teacher-evaluation protocol. Where stochastic, repeat with several frozen seeds and report variation. Where a method produces no meaningful calibrated confidence, compare unscored proposals or ranking separately rather than faking probability outputs.

Useful sampler ablations include: initial-scene sampling only versus dynamic branch-resampling; unique destinations versus raw samples; learned confidence threshold on/off; diverse sampling versus top-likelihood-only; 2D cell sampling versus 3D continuous pose sampling; and performance across

varying placement distances and clutter. Report total sampling **time**, not just number of sampled destinations.

15.3 X/Y learning comparisons

Main model	Purpose	Matched condition
D: coupled X/Y	Primary joint two-outcome representation with contextual quadratic interaction	Same roots, X/Y labels, candidate pool and search budget as E
E: uncoupled X/Y	Tests value of factorization without quadratic interaction	$w_r = 0$ with matched capacity/compute as feasible
DIRECT-GEO	Direct combined-success predictor control	Same root and candidate evidence
G: order-invariant X/Y	Tests whether ordered-prefix features matter	Same endpoints, equal order-opportunity coverage
Geometry/analytic selector	Nonlearning placement/repair selection control	Disclose task-specific information and identical proposal access

In the earlier P24 confirmation, order-aware variants had encouraging evidence on difficult K3/K4 examples but no established primary advantage over DIRECT under the intended statistical criterion. P28 showed failure to generalize reliably across unconstrained off-catalogue choices. The final study must therefore include both order-sensitivity evidence and the more demanding **fair factorization-versus-DIRECT** comparison. The factorization is the chosen architecture; its empirical advantage remains a hypothesis.

15.4 End-to-end ablations that isolate scientific value

1. **Historical privileged catalogue versus independent sampler proposals:** quantifies the extent to which earlier results benefited from constructed intended repairs.
2. **Fixed one-shot sampler versus branch-resampling:** tests whether new destinations in edited scenes matter.
3. **Sampler alone/geometry rank versus learned X/Y rank:** isolates whether X/Y adds value beyond sampling likely placeable destinations.
4. **Coupled versus uncoupled versus DIRECT versus order-invariant:** attributes improved success prediction to precise architectural choices, if any.
5. **X-only versus Y-only versus joint X/Y selection:** establishes that the combined event is necessary for successful repairs; these are evaluator ablations, not proposals to drop Y or X from the final method.
6. **Omission and same-endpoint permutation contrasts:** tests intervention attribution and learned order effects.
7. **Travel auxiliary on/off and length/displacement/travel cost terms:** separates success discrimination from economic selection.
8. **No-op versus correction for originally feasible tasks:** measures unnecessary interventions.
9. **Search budget/proposal budget sweeps:** distinguishes model misranking from insufficient candidate coverage or shallow expansion.
10. **2D versus 3D plus held-out layouts:** measures transfer of method semantics, not presumed shared weights.

When using analytic shortcuts that share geometry with a synthetic scene generator, disclose that relationship. A geometry baseline with construction-aware access maps is not the same as a task-agnostic sampler and must not be presented as such.

15.5 Required physical and causal contrasts

The data must contain genuine same-endpoint order contrasts for which AB and BA are physically tested. A learned $X_{AB} \neq X_{BA}$ is only an *order-sensitive prediction* until independent physical outcomes show it reflects the teacher. Likewise, omission-based claims require independently evaluated A, B, AB and AB-minus-one scenarios under the same root/task conventions.

To substantiate intervention-based causal attribution, measure how often the model correctly predicts a counterfactual change in **target outcome Y** when an object is moved, omitted or placed differently, and how often it predicts **X** changes under reordered actions. No explicit causal graph, unique cause identification or object-wise blocker mask is required. Avoid claiming that a nonzero learned w_r identifies a directed causal edge.

15.6 Metrics and outcome statuses

At minimum report: placement sample precision/coverage/diversity; sampler generation time; X/Y root NLL and Brier; probability calibration; complete correction verified success; true no-op accuracy; chosen action count and object displacement; measured joint-space travel when executable; time to first verified repair; teacher calls; proposal count and sequence count; missed positive scenarios; and root-stratified paired uncertainty intervals.

Report execution statuses distinctly:

- **FOUND_VERIFIED:** a candidate was independently verified within the evaluation budget.
- **SELECTED_UNVERIFIED:** predictor chose a candidate, but no independent physical verification was executed; not a success claim.
- **SEARCH_BUDGET_EXHAUSTED_UNKNOWN:** search stopped with unexamined proposals/branches.
- **PROPOSAL_COVERAGE_UNKNOWN:** sampler may not have generated a necessary action; cannot infer no repair.
- **TEACHER_BUDGET_EXHAUSTED / CENSORED:** evaluation could not resolve the case under the pinned physical budget.
- **EXHAUSTED_DECLARED_CANDIDATE_SET:** all enumerated available proposals were checked; does *not* imply the full physical domain is infeasible.

Use the actual tested tool statuses if different, but preserve these semantic distinctions. A prediction that a repair succeeds is not a verified physical success.

15.7 Holdout discipline and anti-leakage

All descendants of a scene root—including hypothetical edited scenes, repeated views, order permutations, sampler labels and teacher witnesses—must remain in the same split. Freeze sampler, X/Y checkpoints, confidence thresholds, cost coefficients, sample count, search limits and seeds using training/validation roots only. Independent confirmation/test roots are untouched until the final evaluation. P24 confirmation records are not a development tuning source. Historical results remain archived, not rewritten to make the new architecture look successful.

15.8 External robotics comparisons and hardware evidence

Where the 2D/3D embodiments and available implementation contracts make them meaningful, the broader benchmark may compare against established rearrangement planners such as M4M, iM4M, eM4M and a clearly scoped RRT* control, and learned/reference methods such as REAP and G-TAMP. **Their names in the planned paper baseline table are not evidence of direct algorithmic comparability.** Each external method must be adapted to the same robot, manipulation skills, initial states, destination support rules, obstacle counts, time/teacher budget and goal definition. Report unsupported capabilities or incomparable assumptions explicitly rather than silently changing the task to favor Approach B.

The paper-level evaluation can group external comparisons into **algorithmic methods** and **learned methods**, with the internal X/Y/DIRECT/G, sampler-family and search-budget ablations reported separately. Hardware should include distinct obstacle/order/destination variants and total execution time, as well as whether the correction and original target action actually succeeded. Record sampler overhead, X/Y inference overhead, low-level controller time, failed execution phases and the number of corrective reattempts; do not report only a favorable aggregate duration.

Hardware results are downstream validation of a high-level predicted correction, not evidence that the high-level scorer generated the continuous robot motion. Negative hardware outcomes and adaptation constraints remain part of the evaluation record.

15.9 Minimum figures for an auditable study

The final paper/technical report should include: (i) a root-to-sampler-to-X/Y architecture diagram, (ii) an original/hypothetical/verified scene comparison for a multi-step correction, (iii) a same-endpoint AB/BA X-contrast with shared Y, (iv) sampler placement maps in 2D and support-pose clouds in 3D with *independently measured* labels, (v) sampler precision/coverage versus proposal count and runtime, (vi) X/Y joint calibration and end-to-end verified repair performance, (vii) failure breakdown into sampler miss, X misranking, Y misranking, finite-budget search and physical teacher/robot failure, and (viii) hardware sequence frames when available. Distinguish schematic, predicted and physically verified visuals in every caption.

16. Implementation scope and migration from present code

16.1 What exists now

The current repository has the source-verified components necessary for much of Approach B's **main model**: versioned X and Y teacher queries; a physical sequential X execution with a terminal connector; hypothetical target-only Y queries from q_{ref}; normalized `bxy_core` probabilities; `bxy_p11`/`bxy_p17` model families and training; and the P26–P28 universal action-space, search and scoring reference. These provide a strong starting implementation, not evidence that the sampler architecture has already been integrated.

The old P20 generator reads construction-informed fields. The P26 enumerator uses all legal currently empty cells without a physical placement sampler. P27 best-first search expands every child of an expanded prefix; P28 scores with frozen ensembles. **None of these currently implements the separate learned placement sampler with dynamic hypothetical-scene resampling specified here.**

16.2 Minimal module boundaries

Maintain a minimal, understandable ownership split rather than a large new framework:

Logical module	Responsibility
`placement_dataset`	New sidecar records and frozen offline placement-label collection on existing roots
`placement_sampler`	cVAE/FM or analytical candidates behind identical `sample(S,o,M)` interface; cache and score provenance
`corrective_search`	Immutable root; branch-local hypothetical edits; candidate append/dedup; queue/budget bookkeeping
`xy_scorer`	Existing normalized X/Y coefficients, marginals, travel readout; whole ordered candidates from S_0
`physical_teacher`	Unchanged, versioned offline X/Y and placement-trial supervision; final independent verification
`evaluation`	Sampler-alone, fixed-candidate X/Y, end-to-end and ablation protocols

These are **ownership boundaries**, not a requirement for six new physical Python files; prefer reusing existing modules and keeping implementation concise. The historical P25 branch, P24 evidence, frozen test results and teacher policy must remain immutable. Add a new development branch/run namespace for the sampler integration. Code changes must be tested against the legacy scoring identity and six archived replays before using new evidence.

16.3 Specific compatibility tests

- The dynamic sampler can be replaced by a deterministic toy sampler without changing X/Y prediction semantics.

- No `construction.intended`, blocker certificate, successful teacher witness, or role-value privilege can alter test-time sampler output when only observable fields are held fixed.
- Different action orders with the same final edited object scene return exactly the same Y score within numerical tolerance, but may return different X scores.
- Appending an additional candidate to the query set does not change an existing Y marginal because of root-consistent normalization.
- Batching or memoizing unchanged sequences does not alter frozen X/Y numerical predictions.
- A failed stopping-point terminal connector does not automatically remove valid extensions of that prefix.
- Dynamic branch updates never mutate the original root or siblings.
- The sampler cannot accidentally relocate an object already moved in the prefix or a fixed/target object under the current grammar.
- Missing and censored placement/X/Y labels remain distinct from observed zeros.
- Sampling probability/density cannot be mislabeled as calibrated feasibility probability without a declared calibration method.

16.4 Development stages within the final architecture

Gate 0 — Data and contract audit. Confirm exact inventory of both 5,000-scene corpora, their current full-repair labels, possible placement-level witness fields, 2D/3D robot/teacher contracts and split identity. Record the missing labels rather than infer them.

Gate 1 — Offline placement labels. Reuse the scene roots, add individual-placement trials, preserve positive/negative/censored outcomes, support/grasp policy identity and edited-scene provenance. Produce visual examples of successful/failed destinations for 2D and 3D.

Gate 2 — Sampler-alone comparison. Implement conditional VAE, conditional flow matching, Sobol and the separately specified swept-volume comparator under one interface. Report calibrated feasibility scores only if justified. Evaluate candidate precision, coverage, collapse, latency and repair-preservation potential.

Gate 3 — Dynamic sampler integration. Replace the old construction-informed catalogue input to the *new development pipeline* with branch-local sampled placements. Preserve X/Y root-relative sequence encoding, Y caching, sound static rules and bounded-search status accounting.

Gate 4 — Matched X/Y training and comparison. Enrich X/Y training with diverse candidate sequences from the same scene roots; compare D/E/DIRECT/G and cost ablations on fixed candidate pools and held-out outcome queries.

Gate 5 — Independent 2D and 3D end-to-end evaluation. Freeze selected hyperparameters, run blind verification, report successes, omissions, order effects, selected displacement and travel, and limitations. No retrospective changing of sample budgets, labels or root rosters after inspecting confirmation outcomes.

The gates are a reproducibility plan, **not** a proposal to revise the finalized architecture. They must not be marked completed without actual runs and artifacts.

17. Failure modes, limitations and truthful claims

17.1 The sampler may miss the only useful destination

A conditional sampler can be efficient yet miss a rare placement that a correct repair requires. This is especially likely if training positives were dominated by short local parking or construction-informed intended destinations. Increasing the number of draws or mixing proposal regions may help coverage, but no finite stochastic sampler has an unrestricted completeness guarantee. Report sampler-induced repair misses separately from X/Y misranking.

17.2 Placement feasibility is not whole-relocation feasibility

The sampler is deliberately conditioned on the object being grasped. It may propose a placeable destination for an object whose initial grasp is blocked, or for which the currently achieved robot pose is incompatible with that placement. This is **not an architectural contradiction**. X predicts whether the full ordered sequence can realize the object placements, including valid picking, carry, release, transitions and terminal connector.

The grasp-agnostic placement convention must nonetheless be specified in the label protocol. A claim of placement feasibility for "some" permitted grasp cannot be interpreted as a guarantee for an arbitrary grasp; the X teacher searches or uses allowed grasp/motion realizations under a fixed policy.

17.3 Dynamic hypothetical states are not actual achieved robot states

S^r represents object poses only. It is used to generate future placement proposals and Y scenes. The actual robot configuration after a prefix is unknown to this object-only sampler. The ordered X predictor uses sequence features to estimate the effects of this implicit state dependence; it does not guarantee a valid trajectory. Independent teacher execution remains essential for scientific validation and real robot control.

17.4 Energy, confidence and the optimizer's curse

The joint Boltzmann model is normalized and internally consistent, but can still be miscalibrated on candidates far outside its training distribution. Optimizing its score over many generated alternatives can amplify prediction errors. Ensemble disagreement, out-of-distribution distance, calibration and verification are therefore reported. None constitutes an implicit extra learned causal head. The sampler's own generative density is also not a substitute for calibrated placement success probability.

17.5 No global infeasibility theorem

Failure to find a repair may stem from missing sampler candidates, insufficient sequence depth, shallow search, incorrect X/Y probabilities, a constrained action/skill grammar or bounded teacher execution. Such a result is **unknown within budget**, not a proof that a repair does not exist. Only independently established sound geometric or topological certificates justify stronger statements, with their assumptions and scope disclosed. Such certificates are diagnostic/baseline tools, not privileged inputs defining the primary learned sampler's allowed repair candidates.

17.6 No implicit claim of explicit causal identification

The term *causal* is justified by controlled interventions and counterfactual comparison under a fixed scene/task protocol. It does not mean the factor graph is a causal DAG, the $w_r xy$ term reveals a directed cause, or the model outputs a unique minimum blocker set. Strong claims require experimentally validated outcomes under omitted, changed and reversed corrections.

17.7 Evidence boundaries and reproducibility

The P26–P28 source and reports establish that universal grammar/search and frozen scoring have been implemented and tested on the declared 2D development cases; they do **not** demonstrate final sampler-conditioned, complete physical repair success. The 5,000-scene datasets are reported as in progress; their exact data fields require audit. Nothing in this final methodology turns a proposal, visualization or predicted success into a verified robot execution.

18. Frozen decision record and what remains open

18.1 Architectural decisions: final

Decision	Final requirement
Proposal object	`{object, relocate, explicit destination pose}`; sampler supplies placements only.
Sampler semantics	Predict plausible robot-placeable destinations **assuming grasp success**; no target repair identification.
Candidate set	Constructed from sampler proposals; branch-specific dynamic resampling permitted and part of intended search mechanism.
Main learned system	Existing two-outcome X/Y Boltzmann architecture; no new intermediate motion variable or explicit causal graph.

Decision	Final requirement
X	Complete ordered physical relocation sequence with achieved-state carry-over, plus final reference-state connector.
Y	Original target task in final hypothetical edited scene from fixed reference state, independent of relocation ordering.
Selection	Joint X/Y success plus declared length/displacement/travel costs; no hidden physically verified candidate shortlist.
Training data	Reuse stated 5,000 2D and 5,000 3D roots; add placement labels as sidecars.
Test-time feasibility	No IK/teacher query for every sampler output before ranking; external execution/evaluation independent.
Scientific validity	Matched sampler-alone and X/Y/end-to-end evaluation with no solution-construction leakage.

18.2 Implementation/ablation decisions: legitimately open

- Choose **conditional VAE or conditional flow matching** for the selected learned sampler, using controlled sampler-alone evidence; Sobol and swept-volume alternatives remain comparators. The swept-volume proposal mechanism must be supplied and audited before it can be specified precisely.
- Select the grasp-agnostic **offline label aggregation** convention (e.g., attempted placement success under a pinned allowed-grasp policy or existence among a finite fixed grasp set), and pin the precise meaning of calibrated confidence.
- Fix labelled-trial budgets within the existing scene roots, whether sampler confidence uses calibration/readout, and how hypothetical intermediate states are selected for labels. These are measurement/implementation decisions, not changes to sampler responsibility.
- Choose per-object destination budget, branch-expansion schedule, maximum search horizon appropriate to each embodiment, no-op threshold and selection penalties using development-only evidence.
- Determine 2D versus 3D encoder/pose parameterizations and whether any weights are shared; this does not change the two-component architecture.

There is no pending decision to add a learned sequence generator, explicit causal graph, predicted joint-state transitions, target-specific intended-repair sampler, or a mandatory online physical verifier. Those would be substantive architectural deviations and are outside this frozen design.

18.3 The definitive end-to-end statement

Final-fixed Approach B: In a blocked manipulation scene, a placement-conditioned sampler proposes explicit destination poses for movable obstacles, assuming a viable grasp and without evaluating the target-repair objective. A bounded branch-specific search forms ordered corrective sequences and may resample placements after hypothetical object-state edits. A root-consistent, prefix-conditioned Boltzmann X/Y model then predicts (X) whether each entire ordered relocation sequence and its terminal target-reference connector can physically execute, with no intermediate reset, and (Y) whether its resulting hypothetical object arrangement enables the original target task from a fixed reference robot state. The model ranks complete corrections by normalized joint-success probability and correction costs. It does not generate placements or robot motion, learn an explicit causal DAG, or assume intermediate robot configurations; offline physical experiments provide independent supervision and evaluation. The scene corpus is reused, while new placement-level labels are collected within those scenes. This architectural division is fixed; sampler-family selection and computational budgets are controlled experiments within it.

Appendix A. Minimal typed records and interface examples

The following records are conceptual contracts, not a claim that these exact keys already exist in repository source.

A.1 Runtime sampler request

```
{
  "root_id": "scene_2d_0001",
  "scene_view": "hypothetical object poses after frozen prefix",
  "object_id": "A",
  "conditioning": "grasp-success / declared grasp-agnostic placement policy",
  "support_domain": "legal grid cells or 3D support poses",
  "proposal_budget": 8,
  "sampler_version": "sample-model-or-analytic-method-v1"
}
```

A.2 Runtime sampler response

```
{
  "proposals": [
    {
      "object_id": "A",
      "action": "relocate",
      "destination": "cell_23_or_SE3_pose",
      "proposal_score": 0.82,
      "score_semantics": "uncalibrated_density_or_calibrated_p_place",
      "sampler_id": "cvae_v1"
    }
  ],
  "model_runtime_ms": null,
  "static_eligibility_policy": "F1_legality_v1"
}
```

A number such as `0.82` above is deliberately **not** called a calibrated feasibility probability unless the named scoring implementation and supporting calibration evidence make that interpretation valid.

A.3 Complete sequence passed to X/Y

```
{
  "original_root": "scene_2d_0001",
  "original_robot_state": "q0",
  "reference_robot_state": "q_ref",
  "ordered_relocations": [
    {"object_id": "A", "destination": "P"},
    {"object_id": "B", "destination": "Q"}
  ],
  "endpoint_scene": "Edit(S0, [(A,P), (B,Q)])",
  "sampler_provenance": ["branch_S0", "branch_after_A"]
}
```

The scorer returns p_Y, $p_{X|Y}$, p_{succ}, optionally marginal p_X and $\widehat{J}$, and selection cost. The values are predictions, not test-time truth labels.

Appendix B. Required conceptual regression examples

1. **AB versus BA:** Same final poses, same Y key; distinct X keys; teacher contrasts test genuine order effects.
2. **A insufficient, AB sufficient:** $Y_A = 0$ but $Y_{AB} = 1$; A is not wrongly pruned for low Y.
3. **Terminal connector trap:** A's physical relocation succeeds but stopping after A cannot connect to q_{ref}; AB may still be physically successful. Do not conflate stopped-prefix X with relocation-only viability.

4. **B initially inaccessible:** The sampler may propose placement of B conditional on grasp; X rejects B-first if its pick fails; moving A may enable AB. Dynamic proposals can still change B's possible destination slots.
5. **Sampler false positive:** A destination is predicted placeable but execution fails; X can learn that whole-sequence outcome is unfavorable, and sampler-alone precision records the error.
6. **Sampler false negative/miss:** The only necessary destination is absent from sampled proposals. Even perfect X/Y cannot select it. Count this as a coverage failure.
7. **No-op scene:** Original target already feasible. The learned selector does not impose unnecessary object motions.
8. **Harmful relocation:** Placement succeeds individually, but leaves target infeasible or creates new interference. Sampler should not be blamed for Y; main model should distinguish the failure.
9. **Different orders, identical Y:** Appending extra X queries that share one Y scene does not change Y's normalized marginal.
10. **2D-to-3D semantic parity:** Although placement pose dimension changes, the X/Y event definitions and data provenance remain invariant.

Appendix C. Mathematical quantity glossary

Symbol	Meaning	Learned or fixed?
S_0, q_0, q_{ref}	Original object scene, initial robot state, target-reference robot state	Observed/fixed root input
$o, d, a = (o, d)$	Movable object, destination, relocation action	Supplied / sampled explicit inputs
$r = (a_1, \dots, a_m)$	Complete ordered corrective prefix	Search-constructed input
S^r	Hypothetical final object scene after editing r	Deterministic object-pose edit
V_{place}	Individual placement outcome conditional on prescribed post-grasp policy	Offline teacher label / sampler prediction target
X_r	Ordered corrective execution plus connector outcome	Offline teacher binary / predicted by main model
$Y_{s(r)}$	Original target feasibility in edited scene from q_{ref}	Offline teacher binary / predicted by main model
a_r, b_s, w_r	Neural X-unary, Y-unary and X–Y coupling logits	Learned main-model coefficients
$p_Y, p_{X\|Y}, p_{\text{succ}}$	Normalized Y, conditional X and joint success probabilities	Derived from learned coefficients
$\widehat{J}_r$	Predicted joint travel under pinned witness policy	Learned auxiliary readout
$m, C_d, C(r)$	Correction count, object displacement and final decision cost	Deterministic / derived selector
M, K, B	Per-object sampling, search depth and compute/call budgets	Declared controls, not latent variables
λ_J	Travel regression training weight	Validation-tuned
α, β, γ	Selection costs for length, displacement, predicted travel	Validation-tuned

Appendix D. Evidence and revision provenance

This final methodology consolidates the specific user-approved design decisions from the conversation through **5 November 2026** and the following supplied/reviewed technical artifacts:

1. `Approach_B_Final_Methodology_2026-10-31.pdf` — the prior finalized X/Y Boltzmann event definitions, teacher connector contract, F1–F5 specification, normalized root model, observed-data NLL, travel objective, and causal-claim boundaries.
2. The supplied **Methodology v1.7** and **Implementation Guide v1.9** — historical broader project designs and implementation background; where they differ from the final 31 October Approach B and this document, this final document controls **Approach B**.
3. Current repository branch `approach-b-universal-search` at commit `45b0fb05` (full SHA in GitHub), including `bxy_p20_k34.py`, `bxy_f1.py`, `bxy_teacher.py`, `bxy_core.py`, `bxy_p11.py`, `bxy_fullspace.py`, `bxy_fullsearch.py`, `bxy_fullscore.py` and P26–P28 reports `0133`, `0134`, `0135`.
4. P26–P28 are source-evidenced *pre-sampler* implementation/results; no teacher-labelled P29 run or final sampler-trained result is asserted as completed in this document.
5. The reported ongoing 5,000-root 2D and 5,000-root 3D data-generation efforts are **user-specified planned data assets**. An implementation audit must verify the roots and label fields before a report claims actual counts, trials or success rates.

Revision policy: Subsequent documents may refine sampler-family selection, finite budgets, model hyperparameters, physical teacher instrumentation and 2D/3D encodings. A change to the sampler/main-model responsibility split, the two X/Y outcomes, root-relative ordered-sequence evaluation, no intermediate-state variable, no explicit causal graph, or nonprivileged destination proposals is a change to the frozen methodology and must be labelled as such—not silently presented as an implementation detail.